An Analysis of Vanishing gradient problem

Vanishing gradient problem

Vanishing gradient problem -- Part 1

Since | σ ′ | ≤ 1 {\displaystyle \left|\sigma '\right|\leq 1} , the operator norm of the above multiplication is bounded above by ■ W rec ■ k {\displaystyle \left\|W_{\text{rec}}\right\|^{k}} . So if the spectral radius of W rec {\displaystyle W_{\text{rec}}} is γ < 1 {\displaystyle \gamma <1} , then at large k {\displaystyle k} , the above multiplication has operator norm bounded above by γ k → 0 {\displaystyle \gamma ^{k}\to 0} . This is the prototypical vanishing gradient problem. The effect of a vanishing gradient is that the network cannot learn long-range effects. Recall Equation (loss differential): ∇ θ L = ∇ x L (x T , u 1 , … , u T) [∇ θ F (x t − 1 , u t , θ) + ∇ x F (x t − 1 , u t , θ) ∇ θ F (x t − 2 , u t − 1 , θ) + ■] {\displaystyle \nabla _{\theta }L=\nabla _{x}L(x_{T},u_{1},\dots ,u_{T})\left[\nabla _{\theta }F(x_{t-1},u_{t},\theta)+\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{\theta }F(x_{t-2},u_{t-1},\theta)+\cdots \right]} The components of ∇ θ F (x , u , θ) {\displaystyle \nabla _{\theta }F(x,u,\theta)} are just components of σ (x) {\displaystyle \sigma (x)} and u {\displaystyle u} , so if u t , u t − 1 , … {\displaystyle u_{t},u_{t-1},\dots } are bounded, then ■ ∇ θ F (x t − k − 1 , u t − k , θ) ■ {\displaystyle \left\|\nabla _{\theta }F(x_{t-k-1},u_{t-k},\theta)\right\|} is also bounded by some M > 0 {\displaystyle M>0} , and so the terms in ∇ θ L {\displaystyle \nabla _{\theta }L} decay as M γ k {\displaystyle M\gamma ^{k}} . This means that, effectively, ∇ θ L {\displaystyle \nabla _{\theta }L} is affected only by the first O (γ − 1) {\displaystyle O(\gamma ^{-1})} terms in the sum. If γ ≥ 1 {\displaystyle \gamma \geq 1} , the above analysis does not quite work. For the prototypical exploding gradient problem, the next model is clearer.

Vanishing gradient problem -- Part 2

Recurrent network model A generic recurrent network has hidden states h 1 , h 2 , … {\displaystyle h_{1},h_{2},\dots } , inputs u 1 , u 2 , … {\displaystyle u_{1},u_{2},\dots } , and outputs x 1 , x 2 , … {\displaystyle x_{1},x_{2},\dots } . Let it be parameterized by θ {\displaystyle \theta } , so that the system evolves as

Vanishing gradient problem -- Part 3

x t = F (x t − 1 , u t , θ) = W rec σ (x t − 1) + W in u t + b {\displaystyle x_{t}=F(x_{t-1},u_{t},\theta)=W_{\text{rec}}\sigma (x_{t-1})+W_{\text{in}}u_{t}+b} where θ = (W rec , W in) {\displaystyle \theta =(W_{\text{rec}},W_{\text{in}})} is the network parameter, σ {\displaystyle \sigma } is the sigmoid activation function, applied to each vector coordinate separately, and b {\displaystyle b} is the bias vector. Then, ∇ x F (x t − 1 , u t , θ) = W rec diag ■ (σ ′ (x t − 1)) {\displaystyle \nabla _{x}F(x_{t-1},u_{t},\theta)=W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-1}))} , and so

Vanishing gradient problem -- Part 4

d x t = ∇ θ F (x t − 1 , u t , θ) d θ + ∇ x F (x t − 1 , u t , θ) d x t − 1 = ∇ θ F (x t − 1 , u t , θ) d θ + ∇ x F (x t − 1 , u t , θ) [∇ θ F (x t − 2 , u t − 1 , θ) d θ + ∇ x F (x t − 2 , u t − 1 , θ) d x t − 2] ■ = [∇ θ F (x t − 1 , u t , θ) + ∇ x F (x t − 1 , u t , θ) ∇ θ F (x t − 2 , u t − 1 , θ) + ■] d θ {\displaystyle {\begin{aligned}dx_{t}&=\nabla _{\theta }F(x_{t-1},u_{t},\theta)d\theta +\nabla _{x}F(x_{t-1},u_{t},\theta)dx_{t-1}\\&=\nabla _{\theta }F(x_{t-1},u_{t},\theta)d\theta +\nabla _{x}F(x_{t-1},u_{t},\theta)\left[\nabla _{\theta }F(x_{t-2},u_{t-1},\theta)d\theta +\nabla _{x}F(x_{t-2},u_{t-1},\theta)dx_{t-2}\right]\\&\;\;\vdots \\&=\left[\nabla _{\theta }F(x_{t-1},u_{t},\theta)+\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{\theta }F(x_{t-2},u_{t-1},\theta)+\cdots \right]d\theta \end{aligned}}}

Vanishing gradient problem -- Part 5

In machine learning, the vanishing gradient problem is the problem of greatly diverging gradient magnitudes between earlier and later layers encountered when training neural networks with backpropagation. In such methods, neural network weights are updated proportional to their partial derivative of the loss function. As the number of forward propagation steps in a network increases, for instance due to greater network depth, the gradients of earlier weights are calculated with increasingly many multiplications. These multiplications shrink the gradient magnitude. Consequently, the gradients of earlier weights will be exponentially smaller than the gradients of later weights. This difference in gradient magnitude might introduce instability in the training process, slow it, or halt it entirely. For instance, consider the hyperbolic tangent activation function. The gradients of this function are in range [0,1]. The product of repeated multiplication with such gradients decreases exponentially. The inverse problem, when weight gradients at earlier layers get exponentially larger, is called the exploding gradient problem. Backpropagation allowed researchers to train supervised deep artificial neural networks from scratch, initially with little success. Hochreiter's diplom thesis of 1991 formally identified the reason for this failure in the "vanishing gradient problem", which not only affects many-layered feedforward networks, but also recurrent networks. The latter are trained by unfolding them into very deep feedforward networks, where a new layer is created for each time-step of an input sequence processed by the network (the combination of unfolding and backpropagation is termed backpropagation through time).

Vanishing gradient problem -- Part 6

∇ x F (x t − 1 , u t , θ) ∇ x F (x t − 2 , u t − 1 , θ) ■ ∇ x F (x t − k , u t − k + 1 , θ) = W rec diag ■ (σ ′ (x t − 1)) W rec diag ■ (σ ′ (x t − 2)) ■ W rec diag ■ (σ ′ (x t − k)) {\displaystyle {\begin{aligned}&\nabla _{x}F(x_{t-1},u_{t},\theta)\nabla _{x}F(x_{t-2},u_{t-1},\theta)\cdots \nabla _{x}F(x_{t-k},u_{t-k+1},\theta)\\&=W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-1}))W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-2}))\cdots W_{\text{rec}}\operatorname {diag} (\sigma '(x_{t-k}))\end{aligned}}}